Dual-Stage Fourier Neural Operator: Mitigating Spectral Bias for Enhanced Ultrasonic Guided Wave Corrosion Imaging

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***Abstract*—Deep neural networks exhibit spectral bias in data-driven ultrasonic guided wave corrosion imaging, preferentially learning low-frequency components while struggling with high-frequency features critical for defect boundary delineation. This paper proposes a Dual-Stage Fourier Neural Operator (DS-FNO) framework designed to mitigate this limitation. The first stage recovers dominant low-frequency velocity components, while the second stage employs residual learning to capture fine-scale spectral content. Numerical experiments on aluminum plate structures validate the feasibility and effectiveness of the proposed dual-stage architecture for enhanced corrosion imaging.**

Keywords—Fourier Neural Operator, spectral bias, ultrasonic guided wave, corrosion imaging

I. Introduction

Material degradation through corrosion presents significant economic and environmental challenges, with annual costs reaching substantial levels globally [1]. Among various non-destructive evaluation techniques for corrosion detection, ultrasonic guided wave (UGW) imaging has emerged as a particularly effective approach due to its capability for long-range inspection and sensitivity to thickness variations [2].

Recent advances in deep learning have revolutionized UGW-based defect reconstruction by establishing direct mappings between acoustic signals and damage characteristics [3,4]. While deep neural networks (DNNs) have demonstrated promising results in corrosion imaging tasks, they suffer from an inherent limitation known as spectral bias or the frequency principle (F-Principle). This phenomenon manifests as DNNs preferentially learning low-frequency components while struggling to capture high-frequency features during training [5,6]. In the context of defect imaging, this bias often leads to over-smoothed reconstructions that fail to preserve sharp boundaries and fine-scale features critical for accurate damage assessment.

The spectral bias in conventional DNNs may arise from the smoothness properties of gradient-based optimization dynamics [7]. During training, networks converge to low-frequency solutions first, requiring more iterations to learn high-frequency components—if they can be learned at all. While this characteristic may contribute to generalization in many applications, it becomes particularly problematic for UGW imaging where sharp discontinuities at defect boundaries and fine structural details require accurate high-frequency reconstruction.

Recently, a new class of networks known as Neural Operators (NOs) has emerged as powerful surrogate models for solving Partial Differential Equations (PDEs). NOs can learn mappings between infinite-dimensional function spaces, with the Fourier Neural Operator (FNO) serving as a prominent representative [8]. However, even FNOs exhibit the spectral bias phenomenon, struggling with high-frequency features essential for capturing multiscale details [9].

In this work, we propose a Dual-Stage Fourier Neural Operator (DS-FNO) framework specifically designed to mitigate the spectral bias limitation in data-driven UGW corrosion imaging. Our method employs a two-stage end-to-end imaging approach that achieves high-quality corrosion reconstruction by incorporating a complementary second stage to compensate for the inherent frequency bias. The FNO serves as the primary decoder in this process, leveraging its function-space mapping capability to learn the highly nonlinear relationship between input acoustic features and velocity distributions. Section II introduces the forward modeling framework and the DS-FNO architecture. Section III discusses the numerical experiment configuration and inversion results. Section IV concludes the paper.

II. Methodology

This section presents the theoretical foundation and implementation details of the proposed DS-FNO framework for ultrasonic guided wave corrosion imaging. The methodology comprises two key components: first, a forward modeling approach that simulates wave propagation in damaged plates to generate comprehensive training datasets, and second, a dual-stage inversion architecture built upon the FNO.

A. Forward Modeling

To generate training data for corrosion characterization, we employ numerical simulation of guided wave propagation

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in damaged plate structures. In the 2D space-time domain, the governing elastic wave equations for isotropic materials can be formulated as [10]:

$$\boldsymbol{M}(\boldsymbol{x})\ddot{\boldsymbol{u}}(\boldsymbol{x},t)+\boldsymbol{C}(\boldsymbol{x})\dot{\boldsymbol{u}}(\boldsymbol{x},t)+\boldsymbol{S}(\boldsymbol{x})\boldsymbol{u}(\boldsymbol{x},t)=\boldsymbol{F}(\boldsymbol{x},t) \quad (1)$$

where $\boldsymbol{M}(\boldsymbol{x})$, $\boldsymbol{C}(\boldsymbol{x})$, and $\boldsymbol{S}(\boldsymbol{x})$ represent the global mass, damping, and stiffness matrices respectively. The displacement field is denoted by $\boldsymbol{u}(\boldsymbol{x},t)$, while $\boldsymbol{F}(\boldsymbol{x},t)$ indicates the external excitation source.

Transformation to the frequency domain yields [10]:

$$(-\omega^2\boldsymbol{M}+i\omega\boldsymbol{C}+\boldsymbol{S})\tilde{\boldsymbol{u}}(\boldsymbol{x},\omega)=\tilde{\boldsymbol{F}}(\boldsymbol{x},\omega) \quad (2)$$

where ω represents the angular frequency, and $\tilde{\boldsymbol{u}}(\boldsymbol{x},\omega)$ and $\tilde{\boldsymbol{F}}(\boldsymbol{x},\omega)$ are the Fourier transforms of the temporal displacement and source fields respectively. This can be simplified by introducing the complex impedance matrix:

$$\boldsymbol{Z}(\boldsymbol{x},\omega)\tilde{\boldsymbol{u}}(\boldsymbol{x},\omega)=\tilde{\boldsymbol{F}}(\boldsymbol{x},\omega) \quad (3)$$

where $\boldsymbol{Z}(\boldsymbol{x},\omega)=-\omega^2\boldsymbol{M}+i\omega\boldsymbol{C}+\boldsymbol{S}$ represents the frequency-dependent impedance matrix.

The frequency-domain formulation with finite difference discretization provides significant computational advantages, particularly through LU decomposition of the complex impedance matrix, which needs to be performed only once for multiple source configurations [11]. This approach substantially reduces the computational burden for generating the comprehensive dataset required for training.

The relationship between the wavefield distribution $\tilde{\mathbf{u}}$ and the velocity model $\boldsymbol{c}$, which is implicit in the aforementioned equations, can be characterized through a nonlinear forward operator:

$$\tilde{\boldsymbol{u}}=\mathcal{G}(\boldsymbol{c}) \quad (4)$$

where $\mathcal{G}$ denotes the forward modeling operator and $\boldsymbol{c}$ represents the spatial velocity distribution. This nonlinear mapping forms the basis for our inverse problem formulation.

To enable quantitative assessment, we exploit the relationship between velocity and structural thickness. The phase velocity $\boldsymbol{c}_p$ of guided waves at a specific frequency exhibits a dispersive relationship with the plate thickness $\boldsymbol{h}$. This relationship, derived from the Rayleigh-Lamb frequency equations for the selected guided wave mode, can be expressed as:

$$\boldsymbol{h}=\mathcal{H}(\boldsymbol{c}_p,f) \quad (5)$$

where $\mathcal{H}$ denotes the dispersion-based transformation operator and f represents the excitation frequency. This mapping enables direct conversion between velocity maps and thickness distributions, facilitating quantitative corrosion assessment.

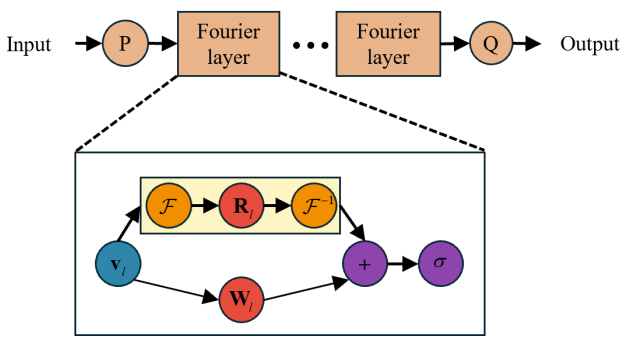


Fig. 1. Fourier Neural Operator architecture with spectral convolution layers.

B. Dual-Stage Inversion Design

The objective of this study is to develop an operator learning framework that establishes robust mappings from acoustic signals to velocity models while mitigating inherent spectral bias. Our proposed DS-FNO approach employs FNO as the decoder in a two-stage process: the first stage primarily recovers low-frequency components of the velocity models, while the second stage specifically targets the fine-scale spectral content that remains under-resolved in the initial reconstruction. By combining outputs from both stages, DS-FNO effectively recovers a broader spectrum of the inverted models, resulting in enhanced reconstruction accuracy.

The FNO architecture, illustrated in Fig. 1, leverages the convolution theorem in Fourier space to efficiently learn operator mappings. The architecture consists primarily of multiple Fourier layers that perform spectral convolutions. Each FNO layer can be expressed as:

$$\boldsymbol{v}_{l+1}(\boldsymbol{x})=\sigma\left(\boldsymbol{W}_l\boldsymbol{v}_l(\boldsymbol{x})+\mathcal{K}_l(\boldsymbol{v}_l)(\boldsymbol{x})\right) \quad (6)$$

where $\boldsymbol{v}_l$ denotes the l-th layer features, σ is a nonlinear activation function, $\boldsymbol{W}_l$ represents linear transformations, and $\mathcal{K}_l$ is the kernel integral operator parameterized in Fourier space [8]:

$$\mathcal{K}_l(\boldsymbol{v}_l)(\boldsymbol{x})=\mathcal{F}^{-1}\left(\boldsymbol{R}_l\cdot\mathcal{F}(\boldsymbol{v}_l)\right)(\boldsymbol{x}) \quad (7)$$

Here, $\mathcal{F}$ denotes the Fourier transform, and $\boldsymbol{R}_l$ is a learnable complex-valued weight tensor that operates on the predetermined lowest Fourier modes, enabling feature mixing and transformation in the spectral domain.

The DS-FNO framework architecture is illustrated in Fig. 2, where both the Stage 1 and Stage 2 models employ four FNO layers with identical architectural configurations as their primary decoders. The two-stage reconstruction workflow proceeds as follows:

The first stage network $\mathcal{N}_1$ maps the input acoustic signals $\boldsymbol{a}$ to an initial velocity reconstruction $\hat{\boldsymbol{c}}_1$:

$$\hat{\boldsymbol{c}}_1=\mathcal{N}_1(\boldsymbol{a};\boldsymbol{\theta}_1) \quad (8)$$

where $\boldsymbol{\theta}_1$ represents the trainable parameters of the first stage. Due to the inherent spectral characteristics of the FNO architecture, this initial reconstruction naturally emphasizes

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the dominant low-frequency components of the velocity distribution.

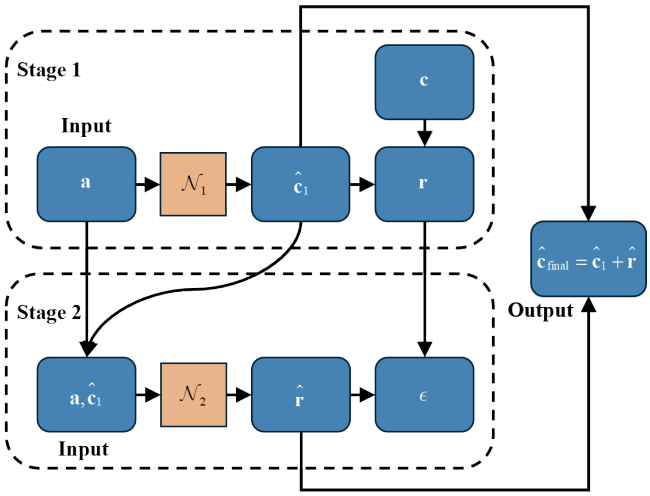


Fig. 2. Dual-stage FNO framework architecture.

The second stage addresses the reconstruction errors by learning a complementary mapping. We define the residual as:

$$\boldsymbol{r} = \boldsymbol{c} - \hat{\boldsymbol{c}}_1 \tag{9}$$

where $\boldsymbol{r}$ represents the residual between the true velocity map $\boldsymbol{c}$ and the Stage 1 prediction, serving as the target for Stage 2 training. The Stage 2 network $\mathcal{N}_2$ takes both the original signals and the Stage 1 output as inputs:

$$\hat{\boldsymbol{r}} = \mathcal{N}_2([\boldsymbol{a}, \hat{\boldsymbol{c}}_1]; \boldsymbol{\theta}_2) \tag{10}$$

where $\boldsymbol{\theta}_2$ represents the trainable parameters of the second stage, and $\hat{\boldsymbol{r}}$ denotes the predicted residual correction. The concatenation $[\boldsymbol{a}, \hat{\boldsymbol{c}}_1]$ provides the second stage with both the original signals and the current reconstruction state, enabling focused learning on the under-resolved spectral components. To enhance the information flow between stages, we incorporate a learnable forward network that transforms the Stage 1 output before concatenation, allowing adaptive feature extraction for residual prediction.

The final velocity map $\hat{\boldsymbol{c}}_{\text{final}}$ combines both stage outputs:

$$\hat{\boldsymbol{c}}_{\text{final}} = \hat{\boldsymbol{c}}_1 + \hat{\boldsymbol{r}} \tag{11}$$

III. Numerical Validation and Results

A. Experimental Configuration

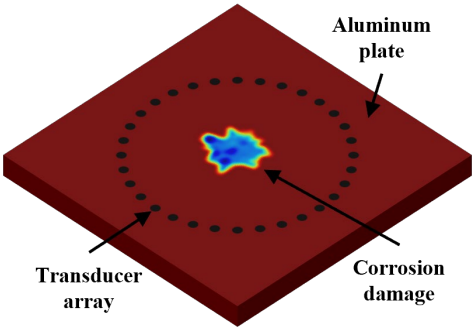


Fig. 3. Experimental setup with aluminum plate and circular transducer array configuration.

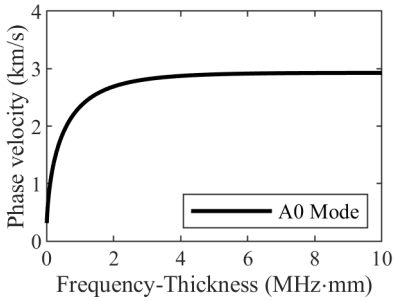


Fig. 4. Dispersion curves for A0 mode in aluminum plate.

The numerical experiments were conducted on an aluminum plate model to validate the proposed DS-FNO framework for corrosion imaging. The simulation setup, illustrated in Fig. 3, consists of a 1000 mm × 1000 mm × 10 mm aluminum plate with material properties: density 2.7 g/cm³, Young's modulus 70.8 GPa, and Poisson's ratio 0.33. A circular array of 32 transducers with a diameter of 0.7 m was uniformly distributed to provide comprehensive wavefield coverage. Various irregular-shaped corrosion defects ranging from 0 to 3 mm in depth were randomly positioned within the transducer array to generate diverse training scenarios. The computational domain was discretized into a 250 × 250 grid with 4 mm spacing. For excitation, a 50 kHz Ricker wavelet was employed at each transducer location sequentially, while the remaining transducers recorded the scattered wavefields. The frequency-domain forward modeling directly provided complex-valued wavefield data, from which both real and imaginary components were extracted as network inputs, resulting in a 2 × 32 × 32 dimensional input tensor for each sample. The corresponding velocity maps served as ground truth labels for supervised training.

Fig. 4 presents the dispersion curves calculated for the aluminum plate with the aforementioned material properties. The A0 mode, when excited at 50 kHz, exhibits strong sensitivity to thickness variations, making it suitable for corrosion depth characterization. Fig. 5 illustrates a representative example of the thickness-to-velocity conversion, demonstrating the monotonic relationship between corrosion depth and phase velocity that enables quantitative damage assessment through the dispersion-based transformation described in (5).

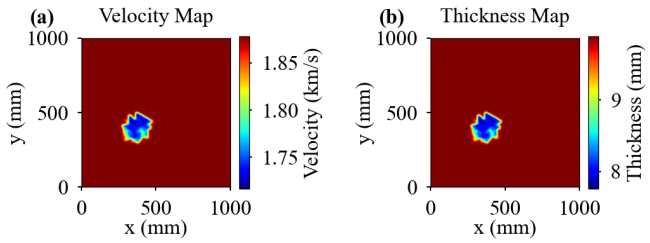


Fig. 5. Thickness-to-velocity conversion example.

B. Results Analysis

The DS-FNO framework was trained and evaluated on a comprehensive dataset comprising 2000 samples generated through the forward modeling process described above. The dataset was partitioned into training, validation, and test sets

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with a ratio of 1400:300:300, ensuring sufficient diversity for model generalization. Importantly, the sample allocation remained consistent across both training stages to maintain fair comparison and prevent data leakage between stages.

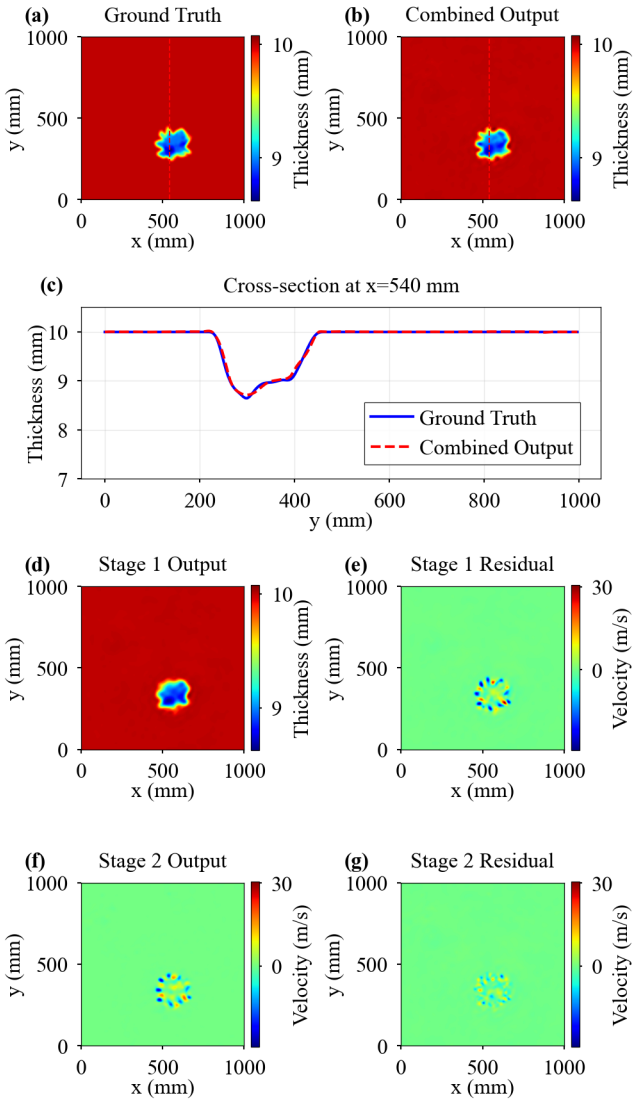


Fig. 6. Inversion results for a representative test sample.

Both stages of the DS-FNO were trained with identical hyperparameters, employing the same optimizer and normalized mean squared error (MSE) loss function. Each stage was trained with early stopping based on validation loss to prevent overfitting. The first stage was trained independently to convergence, after which its weights were frozen during the second stage training to focus exclusively on residual learning.

Fig. 6 presents a reconstruction example from the test set, demonstrating the progressive improvement achieved through the dual-stage architecture. Fig. 6(a) shows the ground truth thickness map with an irregular corrosion defect, while Fig. 6(b) displays the final combined output from the DS-FNO framework, showing excellent agreement with the ground truth. Fig. 6(c) provides a quantitative cross-sectional comparison at a randomly selected location (x=540 mm), where the combined output (dashed line) closely follows the ground truth profile (solid line), accurately capturing both the defect depth and its spatial extent. Fig. 6(d) presents the Stage 1 output, which successfully identifies the defect location but exhibits over-smoothed boundaries. Fig. 6(e) shows the actual residual between the ground truth and Stage 1 output, revealing the high-frequency components that need to be recovered. Figs. 6(f) and 6(g) display the Stage 2 output and its corresponding residual, demonstrating how the second stage compensates for the high-frequency components missed in the initial reconstruction, effectively capturing fine-scale features at defect boundaries and validating the dual-stage approach.

IV. Conclusion

This work presents a Dual-Stage Fourier Neural Operator framework designed to address the spectral bias inherent in DNN-based ultrasonic guided wave corrosion imaging. By employing a two-stage architecture, the method first utilizes FNO to capture dominant low-frequency components, then applies residual learning to recover high-frequency details. Experimental validation demonstrates the framework's ability to achieve high-quality corrosion reconstruction.

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